Project MO V1
Supplier Print Brief

Eight-piece dimensional fit-test coupon set | Revision 1.0

Document	Project MO V1 Supplier Print Brief
Client	Heitor de Moraes Fagundes
Issue date	14 September 2026
Scope	Quotation and manufacture of one complete eight-coupon set
Source format	3MF model files; no G-code supplied

Supplier action requested

- Review every model and the requirements in this brief.
- Return a written quotation and identify the proposed printer, material, settings and lead time.
- Raise manufacturability concerns before changing, repairing, rotating or rescaling any model.
- After approval, print one of each coupon and preserve identification throughout production.

Required response	Supplier entry
Company / contact	
Quotation reference	
Currency and total price	
Lead time	
Shipping cost and destination terms	
Quotation validity	

1. Production requirements

These coupons will be used to measure real printer and material behavior before the full enclosure is released.

Provisional process preferences

Parameter	Requested / supplier response
Process	FDM/FFF unless an alternative is proposed and approved
Material	PETG preferred; state exact brand, grade and color
Layer height	0.20 mm provisional
Infill	20 percent provisional
Supports	Avoid where possible; identify every model that requires support
Quantity	One complete set: one of each of eight coupons
Units and scale	Millimetres, 1:1; do not rescale
Geometry changes	None without written approval
G-code	Supplier-generated only after printer/profile confirmation

Supplier must declare

- Printer manufacturer and exact model.
- Nozzle diameter, layer height, wall/perimeter count, top/bottom layers and infill pattern.
- Material manufacturer, product name, color and whether the filament is dry/conditioned.
- Model orientation and any support, brim or other adhesion strategy.
- Slicer name/version and any compensation, repair or scaling operation.

Quality conditions

- No visible layer shift, severe warping, delamination or incomplete walls.
- Openings, slots, pilots and text must remain clear and identifiable.
- Remove only routine support and brim material; do not sand or enlarge fit features unless specifically authorized.
- Package parts to prevent heat or impact deformation during shipment.

2. Model manifest

Calculated mesh bounds are supplied as a scale-screening reference. Orientation axes may differ from the supplier build orientation.

Item	File	Mesh bounds (mm)
Amplifier	Project_MO_V1_Amplifier_Fit_Coupon.3mf	30.00 x 30.00 x 6.00
ESP32	Project_MO_V1_ESP32_Fit_Coupon.3mf	18.00 x 66.00 x 40.00
Fastener	Project_MO_V1_Fastener_Fit_Coupon.3mf	80.00 x 35.00 x 13.00
Limb interface	Project_MO_V1_Limb_Interface_Fit_Coupon.3mf	42.00 x 42.00 x 6.00
Microphone	Project_MO_V1_Microphone_Fit_Coupon.3mf	35.00 x 25.00 x 8.00
Speaker	Project_MO_V1_Speaker_Fit_Coupon.3mf	60.00 x 60.00 x 21.00
Switch	Project_MO_V1_Switch_Fit_Coupon.3mf	45.00 x 45.00 x 3.00
Dedicated USB-C	Project_MO_V1_USB_C_Fit_Coupon.3mf	30.00 x 25.00 x 25.00

Interface notes

- Dedicated USB-C coupon opening: 9.60 x 3.80 mm with R1.80 corners.
- ESP32 coupon USB opening: 10 x 5 mm with R1 corners. This is a different interface.
- Amplifier guide is intentionally open on one side; omitted central steps did not overlap the board.
- Speaker cradle uses a 10 mm full ring plus two additional 8 mm semicircular sections; updated side distances are 19.0 mm and 16.5 mm.

Digital integrity

All eight 3MF containers passed archive-integrity checks and contained readable mesh data and embedded thumbnails. SHA-256 hashes and exact byte sizes are provided in the CSV manifest.

3. Quotation and pre-production response

Question	Supplier response
Can all eight 3MF files be printed at 1:1 scale without geometry repair?	
Proposed printer and nozzle	
Proposed PETG or alternative material	
Layer height, walls, top/bottom layers and infill	
Orientation and support plan by coupon	
Price for one complete set	
Price for one repeat set	
Production lead time after approval	
Shipping price and estimated delivery	
Payment method and terms	
Any risks, assumptions or recommended changes	

Approval gate

Production is not authorized merely by receiving these files. The client should approve the quotation, declared process and any proposed changes before printing begins.

Decision	Name / date
[ ] Approved as quoted	
[ ] Approved with written changes	
[ ] Revision requested before production	

4. Delivery and traceability

- Identify each coupon by filename or coupon name.
- Include the final slicer settings or project/profile export when permitted.
- Report unsuccessful prints, reprints and any deviations from the approved process.
- Do not mix parts from different printers, materials or parameter sets without separate labeling.
- Protect thin walls, guides, clips and the speaker cradle during packing.

Delivery record

Field	Supplier entry
Production date	
Printer / machine ID	
Operator	
Material batch / spool ID	
Number of successful parts	
Number of failed or reprinted parts	
Deviations from approved quotation	
Shipment tracking	

Included client test document

Project_MO_V1_Physical_Fit_Test_Plan_Rev1.0.pdf contains the receiving inspection, individual coupon procedures, pass/fail criteria, result sheets and CAD revision rules. The client will perform the fit tests after delivery.

File handling

Treat the 3MF files as the controlled manufacturing geometry for this quotation. If a file appears damaged, ambiguous or incorrectly scaled, stop and request a replacement rather than reconstructing the geometry.

Supplier acknowledgement	Name / date
Files received and readable	
Requirements reviewed	
Quotation attached / reference	